

IEEE AMT TECH DAY 2026

Event report

« Antennas, IoT and Wireless for Development »

13 / 05 / 2026 • Amphithéâtre Aboul Aziz Wane • ESP — UCAD — Dakar



General view of the Aboul Aziz Wane auditorium during the event.



Report prepared by:

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1. Executive summary

On Wednesday, 13 May 2026, the École Supérieure Polytechnique (ESP) of Université Cheikh Anta Diop de Dakar (UCAD) hosted, in the Aboul Aziz Wane auditorium, the first Senegalese and West African edition of the IEEE Antenna & Microwave Tech Day. The event, organized by the Laboratory of Computer Science, Telecommunications and Applications (LITA) in partnership with ASCII, the IEEE Senegal Subsection and the IEEE AP-S and IEEE MTT-S societies, was structured around the theme « Antennas, IoT and Wireless for Development ».

The afternoon programme (14:30–18:30) combined a keynote address, two invited talks, two live demonstrations, a doctoral poster session and an awards ceremony. A total of 62 participants signed the attendance sheet — researchers, doctoral students, undergraduates, engineers and institutional representatives. The panel discussion originally planned could not be held due to time constraints and is rescheduled to the 2027 edition.

This report presents the full proceedings of the event, the scientific and technical contributions, audience feedback and the opportunities opened by this inaugural edition.

Key figures

62	5	7	3	6
registered participants	talks & demos	doctoral posters	prizes awarded	partner institutions

2. Opening ceremony



Opening ceremony — welcome speeches and event presentation.

Opening remarks by Prof. Samuel OUYA — Director of LITA and Rector of UNCHK

On behalf of the Laboratory of Computer Science, Telecommunications and Applications (LITA), Prof. Samuel OUYA — Director of LITA and Rector of the Université Numérique Cheikh Hamidou Kane (UNCHK) — welcomed the participants. He recalled that for over a decade LITA has been one of Senegal's most active research hubs on smart networks, wireless communications, cybersecurity and IoT, and that the IEEE AMT Day fits squarely within the laboratory's mission: to produce, transfer and leverage knowledge for development.

He underlined how vital such an event is for education, research and the Senegalese industrial ecosystem. For students, it is a unique opportunity to interact with experienced researchers and to grasp concretely the applications of antenna and Internet-of-Things technologies. For researchers, it offers a privileged framework for exchanges and partnerships — particularly within IEEE, whose Senegal Subsection is experiencing strong growth.

Prof. OUYA finally thanked the partners (IEEE MTT-S, IEEE AP-S, IEEE Senegal Subsection, ASCII, ESP/UCAD) for their support and acknowledged the commitment of LITA's young researchers in preparing this first edition.

Address by Prof. Khaly TALL — on behalf of the Director of the École Supérieure Polytechnique

Representing the Director of the École Supérieure Polytechnique, Prof. Khaly TALL officially opened the day's activities. He first expressed ESP's pride in hosting, for the first time on the West African continent, this worldwide IEEE MTT-S / IEEE AP-S celebration.

Prof. TALL emphasized the strategic alignment of the event with ESP's mission: to train engineers and researchers able to address Africa's technological challenges — connectivity, smart agriculture, connected health, resource management. He stressed that ESP aims to be a regional centre of excellence open to international cooperation, with IEEE as a flagship partner in this ambition.

He thanked all speakers, panelists and doctoral students for the quality of their preparation, and officially declared the activities of the IEEE AMT Tech Day 2026 open.

3. Program

Time	Activity
14:30 – 14:40	Opening ceremony <i>Prof. Samuel OUYA, Prof. Khaly TALL, Prof. Ibra DIOUM</i>
14:40 – 15:10	Keynote — « From smart antennas to frugal IoT » <i>Prof. Ibra DIOUM</i>
15:10 – 15:30	Talk 1 — « Antennas, IoT and Wireless for Development: deploying a LoRaWAN network at ESP » <i>Dr Pape Abdoulaye FAM</i>
15:30 – 15:50	Talk 2 — « Free-Space Optical communications (FSO): an alternative for the last mile » <i>Dr Mangoné FALL</i>
15:50 – 16:00	Open discussion with speakers
16:00 – 16:15	Coffee break & networking
16:15 – 17:00	Demo 1 — « From sensor to cloud: live end-to-end IoT chain » <i>Dr Ousmane SADIO</i>
17:00 – 17:45	Demo 2 — « Antennas & radiofrequency: measurement and characterization » <i>Dr Mamadou Mansour KHOUMA</i>
17:45 – 18:00	Doctoral poster session & live demos <i>7 doctoral students</i>
18:00 – 18:30	Poster contest awards, closing and networking cocktail

4. Keynote address



Title slide of the keynote address.

Prof. Ibra DIOUM, Deputy Director of LITA, opened the scientific session with a keynote titled « From smart antennas to frugal IoT — What challenges for connected Africa? ». The talk unfolded in six parts: (i) the African challenge in figures; (ii) the evolution of antennas from Hertz's dipole to reconfigurable intelligent surfaces (RIS); (iii) antennas for 5G/6G in the African context; (iv) the philosophy of frugal IoT and LPWAN technologies; (v) high-impact socio-economic applications (agriculture, health, water, energy); (vi) the role of the IEEE ecosystem in Africa and collaboration opportunities.

The keynote highlighted LITA's work and that of its partners — notably reconfigurable superdirective antennas with FSS for IoT Wi-Fi modules, low-cost multi-band patch antennas for 5G/6G, and passive UHF RFID sensors for connected health and the cold chain. It concluded with a call for creating an IEEE Student Chapter at ESP and consolidating the IEEE AP-S / MTT-S Senegal Chapter.

5. Invited talks & demonstrations

Talk 1 — « Antennas, IoT and Wireless for Development »

Dr Pape Abdoulaye FAM • Université Cheikh Anta Diop de Dakar / ESP



Title slide — Dr Pape Abdoulaye FAM (LoRaWAN deployment at ESP).

Dr Pape Abdoulaye FAM shared a hands-on experience of deploying a LoRaWAN network at ESP, illustrating how LPWAN adapts to the African context: low cost, low power, long range (10–15 km), unlicensed spectrum. He detailed the end-to-end architecture — sensors, gateways, network server and application server — together with the chosen technology stack (TTN, ChirpStack).

Several concrete use cases were discussed: precision agriculture (soil moisture, local weather monitoring), water management (groundwater table tracking), energy monitoring of photovoltaic panels. The talk closed on transformation opportunities for Senegalese territories and scaling perspectives.

Talk 2 — « Free-Space Optics (FSO) »

Dr Mangoné FALL • DGI / ESP / UCAD



Communication optique en espace libre (Free Space Optique-FSO) : Une alternative pour le dernier kilomètre

Dr. Mangoné FALL

DGI/ESP/UCAD

Title slide — Dr Mangoné FALL.

Dr Mangoné FALL introduced FSO (Free Space Optics) as a credible alternative for the last mile in non-fibred areas. He addressed the theoretical fundamentals of FSO — laser/LED propagation in free space, link budget equations — together with the specific challenges of tropical climates (rain attenuation, fog, atmospheric turbulence).

Several deployment scenarios were detailed: connecting isolated sites (rural health centres, schools), urban backbones in Dakar, hybrid FSO/RF systems with weather-driven automatic switching, and indoor visible light communications (VLC). The talk paved the way for joint LITA/ESP work on FSO channel characterization in Sahelian conditions.

Demo 1 — « From sensor to cloud »

Dr Ousmane SADIO • Researcher and Lecturer — FST / UCAD

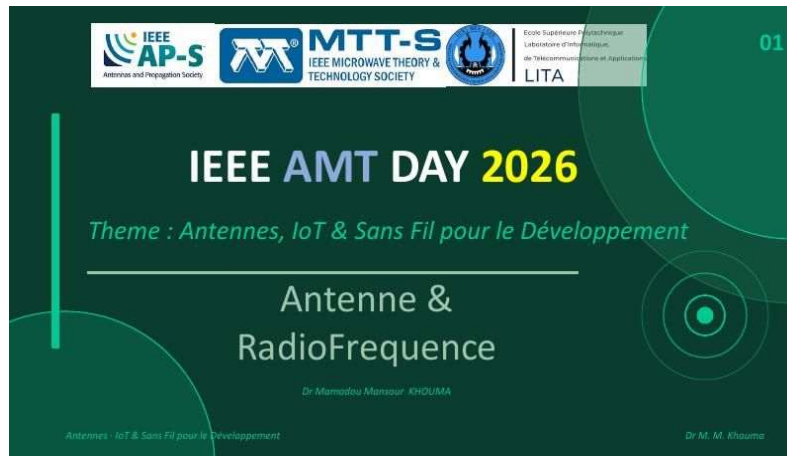
Title slide — end-to-end IoT demonstration.

Dr Ousmane SADIO ran a live, end-to-end IoT chain in under 45 minutes in front of the audience: environmental sensors (temperature, humidity, air quality) wired to an ESP32, LoRa and MQTT transmission, ingestion through a local Mosquitto broker, and real-time visualization in a Grafana / Node-RED dashboard.

Each step was explained and projected on screen: sensor wiring, firmware flashing via Arduino IDE, gateway and broker configuration, final visualization. The audience asked many questions on scalability, transitioning from prototype to field deployment and integration with existing information systems.

Demo 2 — « Antennas & radiofrequency »

Dr Mamadou Mansour KHOUMA • LITA — ESP / UCAD



Title slide — antenna measurement demonstration.

Dr Mamadou Mansour KHOUMA delivered a hands-on demonstration on antenna measurement and characterization. The session covered SOL/SOLT VNA calibration, live S_{11} measurement of patch and PIFA antennas, Smith chart reading and radiation pattern analysis.

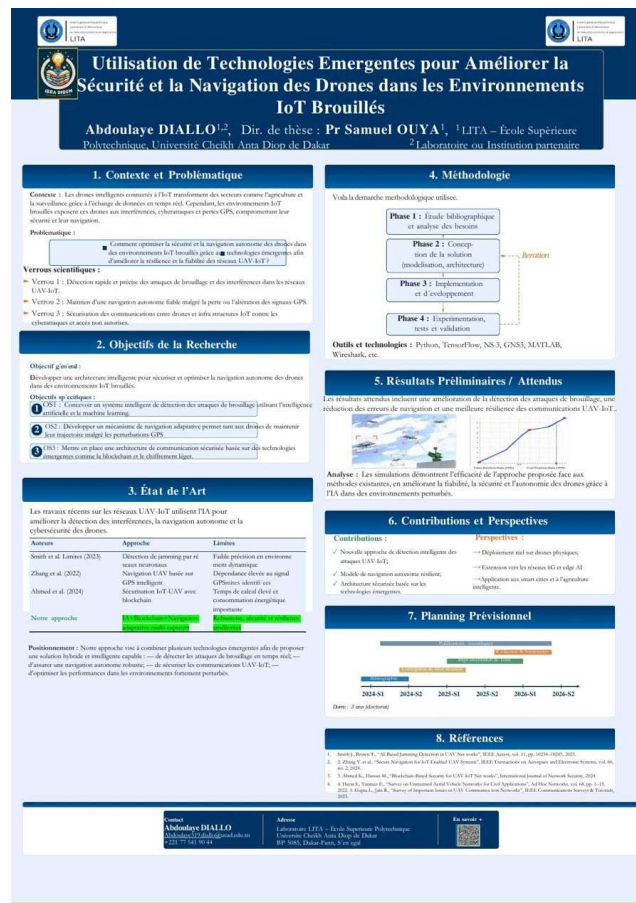
The demo compared a professional VNA (Keysight / Rohde & Schwarz) with a consumer-grade NanoVNA ($\approx$ US\$ 50), illustrating the growing accessibility of RF characterization tools for budget-constrained African laboratories. A discussion followed on the educational value of the NanoVNA for engineering-school lab sessions.

6. Doctoral posters

Seven doctoral students from LITA and partner laboratories presented their work as A0. The jury, composed of two academic researchers and one industry representative, assessed the posters along a weighted six-criterion rubric: scientific clarity, originality, socio-economic impact, demonstration quality, poster graphic quality and oral presentation.

Poster 1 — Drones secured in jammed IoT environments

Abdoulaye DIALLO • PhD supervisor: Prof. Samuel OUYA — LITA / ESP / UCAD



Poster — Abdoulaye DIALLO.

« Using Emerging Technologies to Improve the Security and Navigation of Drones in Jammed IoT Environments ». The work proposes an architecture combining AI, Blockchain and multi-sensor adaptive navigation to ensure UAV resilience against jamming attacks and GPS loss. Announced experiments focus on real-time interference detection and securing inter-drone communications.

Poster 2 — AIoT environmental monitoring

Aché ZAKARIA ALI • PhD supervisor: Prof. Ibra DIOUM — LITA / ESP / UCAD



Poster — Aché ZAKARIA ALI.

« A hybrid, self-adaptive AIoT framework for multimodal environmental monitoring — Towards proactive and resilient ecosystem management in West Africa ». The study designs an Edge / Fog / Cloud architecture coupled with LSTM, CNN and Transformer models, along with an ecological digital twin. Announced targets: multimodal collection (loss < 5 %), self-calibration (error < 5 %), energy optimization (~20 %), digital twin accuracy (+7 %, ≥ 85 %).

Poster 3 — Superdirective antennas with FSS for IoT Wi-Fi

Semou SALL (with M. M. Khouma, I. Dioum) • LITA — ESP / UCAD



Ecole Supérieure Polytechnique
Laboratoire d'Informatique,
de Télécommunications et Applications

LITA



Superdirective Array Antennas with Pattern Radiation Reconfigurable Based on Frequency Selective Surface for IoT Wi-Fi module

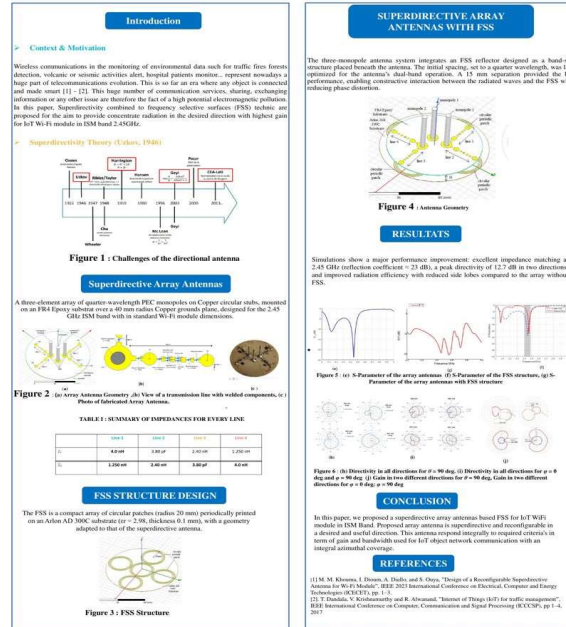
Semou sall, Mamadou Mansour Khouma, Ibra Dioum

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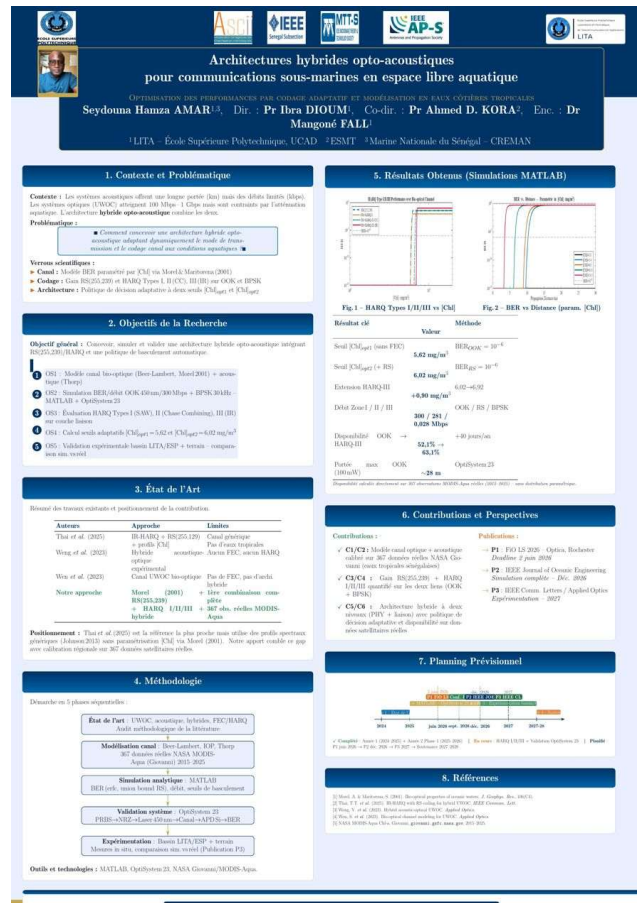


Poster — Semou SALL.

« Superdirective Array Antennas with Pattern Radiation Reconfigurable Based on Frequency Selective Surface for IoT Wi-Fi module ». The poster details the design and simulation of a superdirective antenna array with reconfigurable pattern, based on a frequency-selective surface (FSS) integrated into a 2.45 GHz IoT Wi-Fi module, with significant gain enhancement and simulation-proven directional switching.

Poster 4 — Hybrid opto-acoustic underwater communications

Seydouna Hamza AMAR • Supervisor: Prof. Ibra DIOUM — Co-supervisor: Prof. Ahmed D. KORA — Advisor: Dr Mangoné FALL

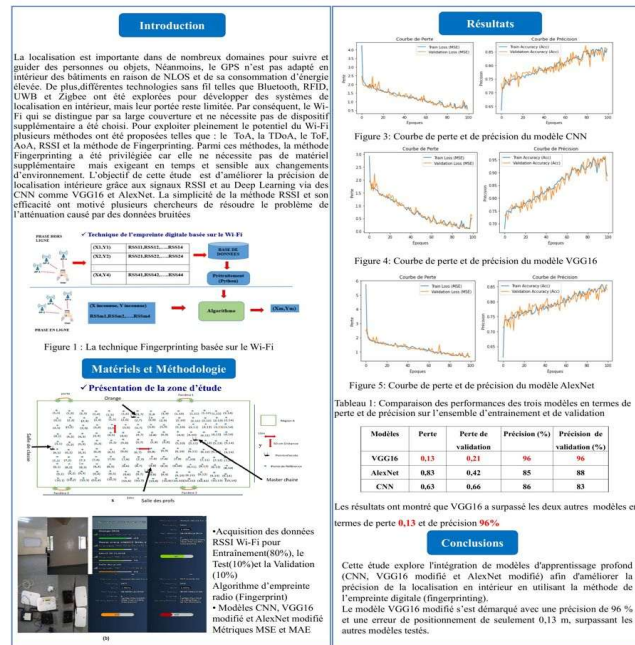


AMÉLIORATION DE LA PRÉCISION DE LA LOCALISATION INTÉRIÈRE BASÉE SUR L'EMPREINTE RSSI À L'AIDE DE CNN, DU VGG16 MODIFIÉ ET DE L'ALEXNET MODIFIÉ

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Département Génie Informatique
École Supérieure Polytechnique (ESP) – Université Cheikh Anta Diop, Dakar-Fann, Dakar 5085, Sénégal.

Mame Maty Gakou, Mariame Niang, Ibra Dioum

Contacts: Mame Maty Gakou, mamematy.gakou@esp.sn; Dr Mariame Niang, mariame.niang@esp.sn



Poster — Mame Maty GAKOU.

« Improving indoor localization accuracy based on RSSI fingerprints using CNN, modified VGG16 and modified AlexNet ». Experiments show that VGG16 outperforms the other models in accuracy: 0.13 m median error and 96 % accuracy within the validation perimeter, demonstrating the impact of deep learning on Wi-Fi-fingerprinting indoor localization.

Poster 6 — Wireless power transfer to passive UHF RFID tags

Demba BA (with Ibra Dioum, Yvan Duroc) • LITA — ESP / Univ. Claude Bernard Lyon 1, INSA Lyon, École Centrale Lyon (CNRS, UMR5505)

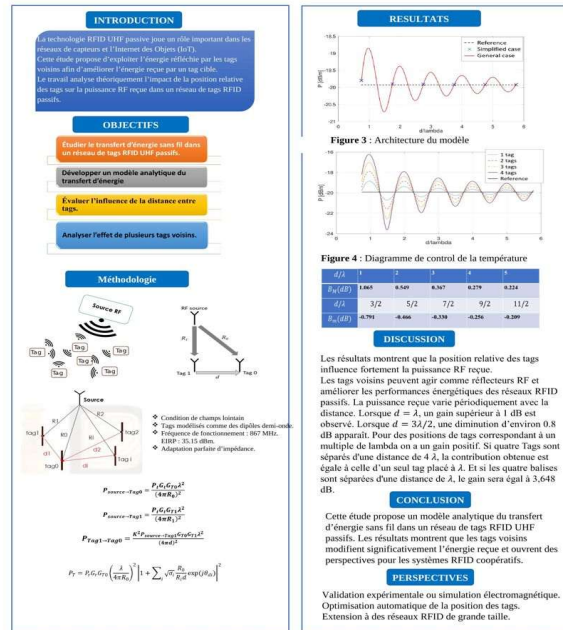


Transfert d'Energie sans fil dans un réseaux de Tags RFID UHF passif

Demba Ba¹, Ibra Dioum¹, Yvan Duroc²

¹LITA - ESP, Dakar, Sénégal

²Université Claude Bernard Lyon 1, INSA Lyon, Ecole centrale de Lyon, CNRS, Ampère, UMR5505, Villeurbanne, France
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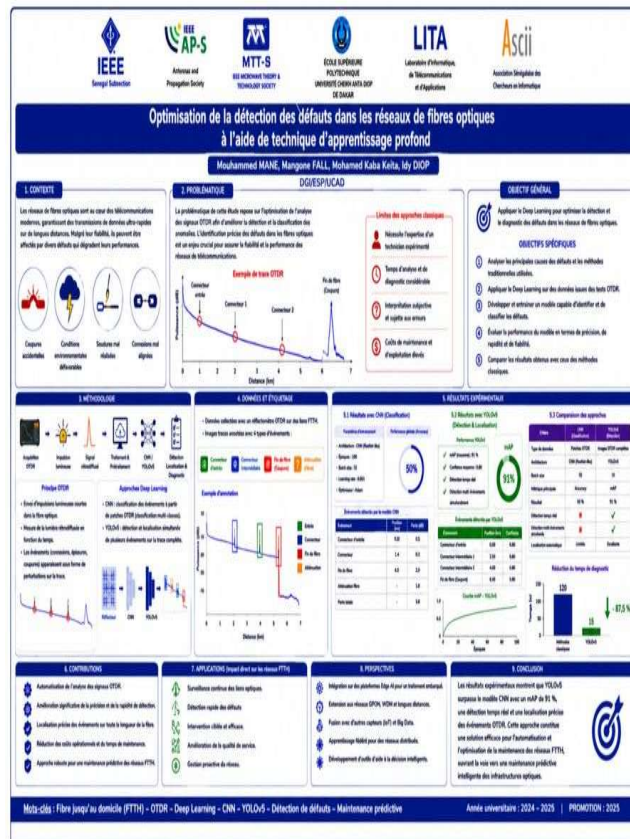


Poster — Demba BA.

« Wireless Power Transfer in a Passive UHF RFID Tag Network ». The poster develops an analytical model quantifying the energy received by a passive tag as a function of distance to the reader and the presence of neighbouring tags. Results show that changes in the reflection coefficient directly affect the received energy. Application: security of coupled RFID chains.

Poster 7 — Optical fibre fault detection by deep learning

Mouhamadou MANE (with Mangoné FALL, Mohamed Kaba Keita, Idy DIOP) • DGI / ESP / ACAO



Poster — Mouhamadou MANE.

« Optimising fault detection in optical-fibre networks using deep-learning techniques ». The poster benchmarks YOLOv5 and CNN on OTDR traces; YOLOv5 reaches 81 % accuracy vs 50 % for a classical CNN. Direct application to predictive maintenance of FTTH networks in Senegal.

7. Awards

Following the jury's deliberation, the following distinctions of the doctoral poster contest were announced at the closing ceremony. The three winners received an IEEE certificate and a prize in front of the audience.

Prize	Awarded work	Awardee
1st prize	« Hybrid opto-acoustic architectures for free-space underwater communications »	Seydouna Hamza AMAR — LITA / ESP / UCAD
2nd prize	« Improving indoor localization accuracy based on RSSI fingerprints using CNN, modified VGG16 and modified AlexNet »	Mame Maty GAKOU — LITA / ESP / UCAD
3rd prize	« Wireless Power Transfer in a Passive UHF RFID Tag Network »	Demba BA — LITA / ESP — Univ. Claude Bernard Lyon 1

8. Participation

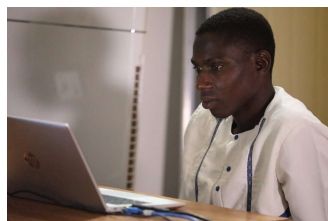
The attendance sheet totalled 62 signatures, distributed as follows: 52 from the pre-registered printed list and 10 walk-in registrations on the day. Most participants were undergraduate and doctoral students from partner institutions (ESP, FST, UCAD, ESMT, IPD, USSEIN, UADB). The professional and institutional sector was represented by engineers (Huawei, Sonatel, A&B Consulting, Codival, ARTP) and researchers (ESP, FST, ESMT, UCAD, UNCHK).

Each registered participant received a unique reference of the form PART-2026-XYZ, which will appear on their IEEE participation certificate.



Attentive audience during the talks session.

9. Photo gallery



10. Conclusion & outlook

This first Senegalese edition of the IEEE Antenna & Microwave Tech Day met its objectives: disseminating advances in antennas, microwaves and IoT to a mixed audience, showcasing doctoral research from LITA and its partners, and fostering a sustainable dialogue between academia, industry and public institutions.

Several opportunities emerge: (i) launch an IEEE Student Chapter at ESP at the start of the 2026 academic year; (ii) consolidate the future IEEE AP-S / MTT-S Chapter in Senegal; (iii) embed the IEEE AMT Day in LITA's annual calendar; (iv) deepen international doctoral collaborations (v) issue a call for papers in a partner technical journal.

11. Acknowledgements

LITA and ESP / UCAD warmly thank Prof. Samuel OUYA and Prof. Khaly TALL for their institutional support and presence at the opening ceremony. We also thank the speakers (Dr Pape Abdoulaye FAM, Dr Mangoné FALL, Dr Ousmane SADIO, Dr Mamadou Mansour KHOUMA), the seven doctoral students who presented posters, the poster-contest jury, and the technical and logistical teams.

We finally thank our institutional partners — IEEE, IEEE MTT-S, IEEE AP-S, IEEE Senegal Subsection, LITA, ESP/UCAD and ASCII — for their support without which this event would not have been possible.

See you for the 2027 edition.